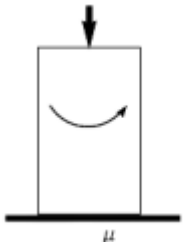
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A solid cylinder of diameter D is resting on a rough horizontal surface as shown in the figure. It exerts a total normal force F uniformly distributed over it's base area. If friction coefficient is μ determine the twisting torque that must be exerted by an external agent to cause Rotatory Motion to impend (about it's symmetry axis)



(A) $\frac{\mu FD}{2}$

(B) $\frac{\mu FD}{4}$

(C) $\frac{\mu FD}{3}$

(D) $\frac{\mu FD}{5}$

Q.2

A wheel starting from rest, is rotating with a constant angular acceleration of $3.0\ \text{rad/sec}^2$. An observer notes that it traces an angle of 120 radian in $4.0\ \text{sec}$ interval. For how long the wheel had been rotated when the observer started his observation?

(A) $4\ \text{sec}$

(B) $2\ \text{sec}$

(C) $8\ \text{sec}$

(D) $6\ \text{sec}$

(E) Solution

Q.3

If the radius of solid sphere is $35\ \text{cm}$. The ratio of radius of gyration, when the axis is along a diameter to that when the axis is along a tangent will be-

(A) $\sqrt{\frac{10}{35}}$

(B) $\sqrt{\frac{35}{10}}$

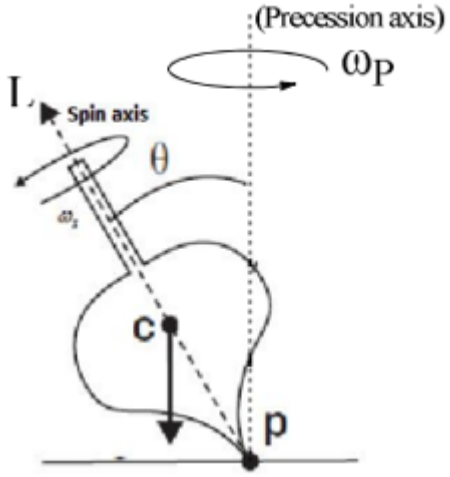
(C) $7/1$

(D) $1/7$

@IITJEE_Advanced

Q.4

Precession is the result of the angular velocity of rotation (spin) and the angular velocity produced by the torque. It is an angular velocity about a line that makes an angle θ with the permanent rotation axis. In the diagram, a top of mass $\mathbf{m}$ is performing precession motion. The torque due to gravitational force causes a change in the angular momentum L in the direction of that torque causing the top to precess with angular velocity ω_{P} which is given by: (Given r is distance of COM from point of rotation ' P '. I_S & ω_{s} are the moment. of inertia and angular velocity of top about spin axis respectively.)



(A) $\frac{3mgr}{I_s\omega_s}$

(B) $\frac{mgr}{2\omega_s I_s}$

(C) $\frac{2mg;}{\omega_s I_s}$

(D) $\frac{mgr}{\omega_s I_s}$

(E) Solution

Q.5

Two circular discs A and B of equal masses and thickness but made of metals with densities d_A and $\mathrm{d_B}$ $(\mathrm{d_A} > \mathrm{d_B})$. If their moments of inertia about an axis passing through the centre and normal to the circular faces be $\mathrm{I_A}$ and $\mathrm{I_B}$, then-

(A) $\mathrm{I_A} = \mathrm{I_B}$

(B) $\mathrm{I_A} > \mathrm{I_B}$

(C) $\mathrm{I_A} < \mathrm{I_B}$

(D) $\mathrm{I_A} \geq \mathrm{I_B}$

Q.6

A particle of mass $3\ \mathrm{kg}$ is moving under the action of a central force whose potential energy is given by $\mathrm{U(r)} = 10\mathrm{r}^3$ joule. For what energy and angular momentum will the orbit be a circle of radius $10\ \mathrm{m}$?

(A) $2.5 \times 10^4\ \mathrm{J}, 3000\ \mathrm{kg\ m^2/sec}$

(B) $2.5 \times 10^3\ \mathrm{J}, 3000\ \mathrm{kg\ m^2/sec}$

@IITJEE_Advanced

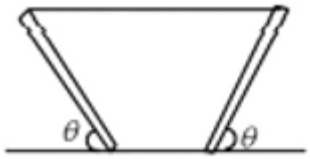
(C) $2.5 \times 10^2\ \text{J}, 30000\ \text{kg m}^2/\text{sec}$

(D) $2.5 \times 10^2\ \text{J}, 300\ \text{kg m}^2/\text{sec}$

(E) Solution

Q.7

Two uniform boards, tied together with the help of a string, are balanced on a surface as shown in figure. The coefficient of static friction between boards and surface is 0.5. the minimum value of θ, for which this type of arrangement is possible is



(A) 30°

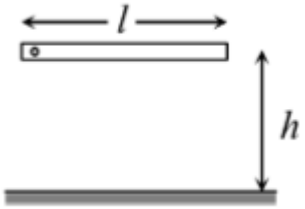
(B) 45°

(C) 37°

(D) It is not possible to have this type of balanced arrangement

Q.8

A thin rod of mass m and length l is hinged at a point which is at a distance $h(h < l)$ above the horizontal surface. The rod is released from rest from the horizontal position. If e is the co-efficient of restitution, the angular velocity of rod just after collision will be $(h = 1\ \text{m}, l = 2\ \text{m}, e = 1)$



(A) $\frac{3\sqrt{3g}}{8}$

(B) $\frac{6\sqrt{3g}}{8}$

(C) $\frac{5\sqrt{3g}}{8}$

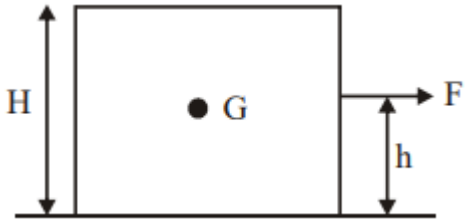
(D) none of these

Q.9

A rectangular block $0.25\ \text{m}$ wide and $0.5\ \text{m}$ high is dragged to the right along a horizontal level surface at constant speed by a force F as shown in figure. The coefficient of sliding friction is 0.4 and the weight of the block is $25\ \text{N}$.

@IITJEE_Advanced

Then which of the following statement/s is/are true?



(A) Force required is $10\ \mathrm{N}$

(B) The value of h at which block just starts to tip is $31\ \mathrm{cm}$.

(C) The shift in line of action of normal reaction from centre of gravity of the block is $5\ \mathrm{cm}$. when $\mathrm{h} = 0.125\ \mathrm{m}$

(D) The force required is $15\ \mathrm{N}$

(E) Solution

^

Q.10

A $0.5\ \mathrm{kg}$ circular disc of radius $10\ \mathrm{cm}$, initially moving with a constant angular speed of $10\mathrm{rad/s}$ is placed coaxially on a stationary disc of mass $1.5\ \mathrm{kg}$ and of radius $10\ \mathrm{cm}$. If the $1.5\ \mathrm{kg}$ disc is placed on a smooth horizontal surface, then -

(A) the final angular speeds of $0.5\ \mathrm{kg}$ and $1.5\ \mathrm{kg}$ discs are respectively in the ratio $1:3$.

(B) both the discs move together with common angular speed of $2.5\mathrm{rad/s}$

(C) no work is done by the friction between two discs.

(D) the fractional loss in kinetic energy is 0.75.

Q.11

A particle is moving in a potential field defined by $\mathrm{V} = -1/\mathrm{r}$. Which of the following statement(s) is/are incorrect in this case?

(A) Angular momentum of the particle is always conserved

(B) Kinetic energy of the particle is always conserved

(C) The particle always follows a closed path

(D) Force on the particle is always radial

(E) Solution

^

Q.12

Passage:

Two boys each of mass $25\ \mathrm{kg}$ are sitting on the opposite ends of a horizontal beam of mass $10\ \mathrm{kg}$ and length $2.6\ \mathrm{m}$. The beam is rotating about a vertical axis through its centre initially at a rate of $5\mathrm{rev/min}$.

Question: Initial angular momentum of the system in $J.s$ is $-$

(A) 20π

@IITJEE_Advanced

(B) 15π

(C) 12π

(D) 25π

Q.13

Passage:
Two boys each of mass $25\ \mathrm{kg}$ are sitting on the opposite ends of a horizontal beam of mass $10\ \mathrm{kg}$ and length $2.6\ \mathrm{m}$. The beam is rotating about a vertical axis through its centre initially at a rate of $5\mathrm{rev/min}$.
Question: What would be the angular velocity if the boys move $0.6\ \mathrm{m}$ towards the centre of the beam without touching the floor?

(A) $15\mathrm{rev/min}$

(B) $10\mathrm{rev/min}$

(C) $25\mathrm{rev/min}$

(D) $20\mathrm{rev/min}$

(E) Solution

^

Q.14

Passage:
Two boys each of mass $25\ \mathrm{kg}$ are sitting on the opposite ends of a horizontal beam of mass $10\ \mathrm{kg}$ and length $2.6\ \mathrm{m}$. The beam is rotating about a vertical axis through its centre initially at a rate of $5\mathrm{rev/min}$.
Question: The change in kinetic energy of the system due to above motion of the boys is -

(A) $22.3\ \mathrm{J}$

(B) $20.1\ \mathrm{J}$

(C) $25.3\ \mathrm{J}$

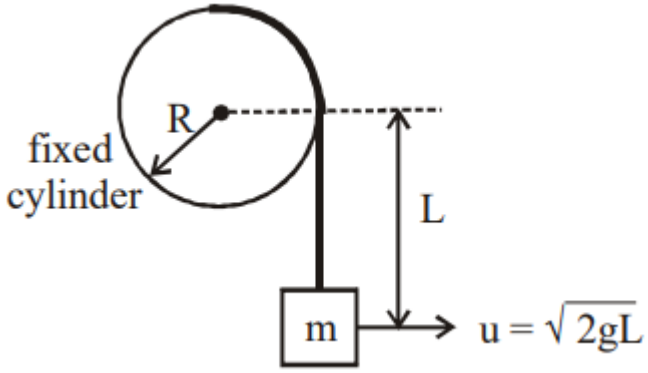
(D) $24.6\ \mathrm{J}$

Q.15

Passage:
The cross-section of a fixed cylinder (not allowed to rotate and translate) with horizontal axis is as shown. One end of a light inelastic string is fixed at top of cylinder of radius R and a small block of mass m is tied to the other end of string. Initially the block is at rest with the portion of string not in contact with cylinder being vertical and having length L as shown. At the lowest position the block is given initial horizontal velocity $\mathrm{u} = \sqrt{2\mathrm{gL}}$ and the block moves in vertical plane. When the block reaches the highest point of its trajectory, the length of string not in contact

@IITJEE_Advanced

with cylinder is $L + \frac{R\pi}{3}$. (where g is acceleration due to gravity).



Question: The distance between block and centre of cylinder when block is at highest position will be -

(A) $2R$

(B) $\sqrt{5}R$

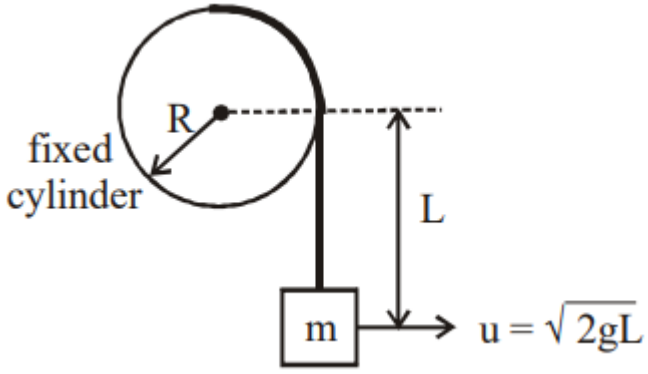
(C) 3R

(D) $2R/\sqrt{3}$

Q.16

Passage:

The cross-section of a fixed cylinder (not allowed to rotate and translate) with horizontal axis is as shown. One end of a light inelastic string is fixed at top of cylinder of radius R and a small block of mass m is tied to the other end of string. Initially the block is at rest with the portion of string not in contact with cylinder being vertical and having length L as shown. At the lowest position the block is given initial horizontal velocity $u = \sqrt{2gL}$ and the block moves in vertical plane. When the block reaches the highest point of its trajectory, the length of string not in contact with cylinder is $L + \frac{R\pi}{3}$. (where g is acceleration due to gravity).



Question: The least tension in string is -

(A) $\frac{mg}{\sqrt{6}}$

(B) $\frac{mg}{\sqrt{5}}$

(C) $\frac{\sqrt{3}}{2}mg$

(D) $\frac{mg}{2}$

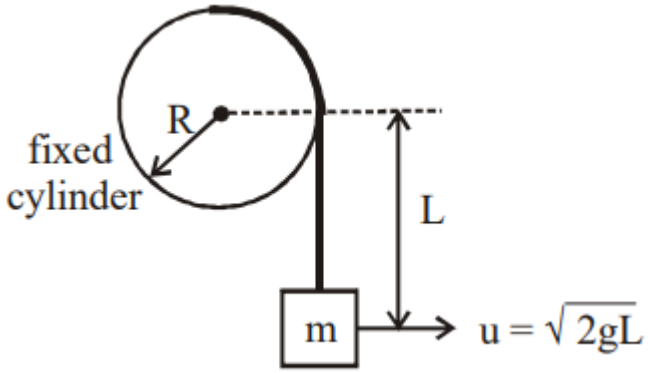
(E) Solution

@IITJEE_Advanced

Q.17

Passage:

The cross-section of a fixed cylinder (not allowed to rotate and translate) with horizontal axis is as shown. One end of a light inelastic string is fixed at top of cylinder of radius R and a small block of mass m is tied to the other end of string. Initially the block is at rest with the portion of string not in contact with cylinder being vertical and having length L as shown. At the lowest position the block is given initial horizontal velocity $\mathrm{u} = \sqrt{2\mathrm{gL}}$ and the block moves in vertical plane. When the block reaches the highest point of its trajectory, the length of string not in contact with cylinder is $L + \frac{R\pi}{3}$. (where g is acceleration due to gravity).



Question: Tangential acceleration of block at highest position is -

(A) $\frac{g}{2}$

(B) $\frac{\sqrt{3}}{2}$ g

(C) $\frac{g}{3}$

(D) $\frac{g}{\sqrt{6}}$

(E) Solution

Q.18

An object is allowed to roll down the incline starting from rest. All are uniform and have same mass and radius.

Column I	Column II
(A) The object which has largest rotational inertia about its axes of symmetry	(p) Solid sphere
(B) The object which will experience the largest net torque	(q) Spherical shell
(C) The object which will have the largest speed at the bottom of the incline	(r) Solid disk
(D) The object which will reach the bottom of incline in the shortest time	(s) Thin hollow cylinder

(A) A-(q) B-(q) C-(p) D-(p)

(B) A-(s) B-(s) C-(p) D-(p)

@IITJEE_Advanced

(C) A-(q) B-(q) C-(r) D-(r)

(D) A-(p) B-(p) C-(q) D-(q)

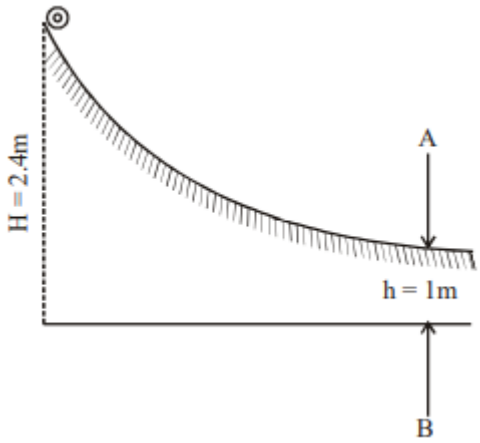
Q.19

A car on a straight road moves with a constant linear acceleration of $7\ \mathrm{m/s^2}$ with a sphere resting on its floor. A man standing outside observes that a sphere is rolling without slipping on the floor of the car. The linear acceleration of the sphere as measured by the man in terms of $\mathrm{ms^{-2}}$ must be.

(B) Solution

Q.20

A small sphere rolls down without slipping from the top of a track in the a vertical plane. The track has an elevated section and a horizontal part. The horizontal part is 1.0 metre above the ground level and the top of the track is 2.4 metres above the ground. Find the distance on the ground with respect to point B (which is vertically below the end of the track as shown in figure), where the sphere lands in terms of metre.



(B) Solution

Answers & Solutions

Q.1 Answer:

$\frac{\mu \mathrm{FD}}{3}$

Solution:

@IITJEE_Advanced

Consider a small element of radius r and width dr.

$$dF = \frac{F}{\frac{\pi}{4}D^2} \times 2\pi r dr$$

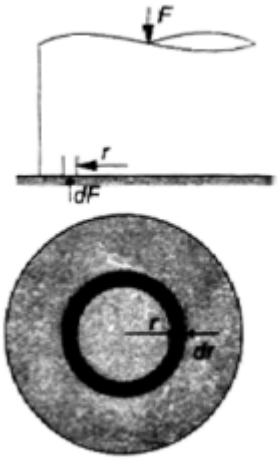
$$dF_f = \mu dF$$

$$dM = r dF_f = r\mu dF$$

$$M = \int dM = \int_0^{D/2} r\mu \times \frac{4F}{\pi D^2} \times 2\pi r dr$$

$$= \frac{4F}{\pi D^2} \times \mu \times 2\pi \times \frac{(D/2)^3}{3}$$

$$M = \frac{\mu F D}{3}$$



Q.2 Answer:

8 sec

Solution:

Let ω_0 be the angular velocity of wheel, when the observer started his observation.

Now, $\theta = \omega_0 t + (1/2)\alpha t^2$

Substituting the values, we get

$120 = \omega_0 \times 4.0 + (1/2) \times 3.0 \times (4.0)^2 \Rightarrow \omega_0 = 24 \text{rad/sec}$

Further, $\omega_0 = \omega_0' + \alpha t$ **.....(1)**

Where ω_0 ' is the initial angular velocity at the starting of motion.

Here $\omega_0 = 24 \text{rad/sec}$

$\omega_0' = 0$ and $\alpha = 3.0 \text{rad/sec}^2$

$\therefore \quad t = 8.0 \text{sec}$ [from (1)]

Q.3 Answer:

$\sqrt{\frac{10}{35}}$

Solution:

@IITJEE_Advanced

Along the diameter, $I_g = (2/5)\mathrm{mR}^2$ **or** $K_g^2 = (2/5)R^2$ **or** $\mathrm{K_g = R\sqrt{\frac{2}{5}} = 35\sqrt{\frac{2}{5}} = 7\ cm}$

Along tangent, $\mathrm{I = Ig + mR^2}$

$\therefore \quad \mathrm{mK^2 = mK_g^2 + mR^2}$ or $\mathrm{K^2 = K_g^2 + R^2 = (2/5)R^2 + R^2 = (7/5)R^2}$

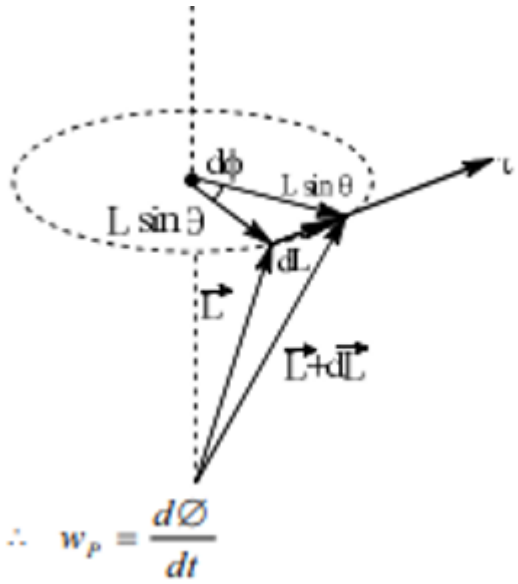
$\mathrm{K = R\sqrt{\frac{7}{5}} = 35\sqrt{\frac{7}{5}}}$, Now $\mathrm{\frac{K_g}{K} = \sqrt{\frac{10}{35}}}$

Q.4 Answer:

$\frac{mgr}{\omega_s I_s}$

Solution:

In time interval dt, $\vec{L}$ rotates through an angle $d\varnothing$



$\therefore \quad w_p = \frac{d\varnothing}{dt}$

$\therefore w_p = \frac{d\varnothing}{dt}$

But $d\varnothing = \frac{dL}{L\sin\theta} = \frac{\tau dt}{L\sin\theta}$

$\therefore \quad \frac{d\varnothing}{dt} = \frac{\tau}{L\sin\theta} = \frac{mgr\sin\theta}{L\sin\theta} \quad = \frac{mgr}{L} = \frac{mgr}{I_s w_s}$

Q.5 Answer:

—

Solution:

$\mathrm{I_A = \frac{m_A r_A^2}{2}}$ **and** $\mathrm{I_B = \frac{m_B r_B^2}{2}}$,

$\therefore \quad \mathrm{\frac{I_A}{I_B} = \frac{r_A^2}{r_B^2}} \quad (\because \mathrm{m_A = m_B}) \ldots\ldots(1)$

Now, $\mathrm{m_A = \pi r_A^2 t d_A}$

$\mathrm{m_B = \pi r_B^2 t d_B}$

So, $\pi r_A^2 t d_A = \pi r_B^2 t d_B$ **or** $\frac{r_A^2}{r_B^2} = \frac{d_B}{d_A} \ldots\ldots\ldots(2)$

From equations (1) and (2), $\mathrm{\frac{I_A}{I_B} = \frac{d_B}{d_A}}$**. As** $\mathrm{d_A > d_B}$ **hence** $\mathrm{I_A < I_B}$

Q.6 Answer:

2.5×10^4 J, 3000 kg m^2/sec

@IITJEE_Advanced

Solution:

The (variable) potential energy of the particle,
$\mathrm{U(r)} = 10\mathrm{r}^3$ **joule.**
Hence the force acting on the particle is $\mathrm{F} = -\frac{\partial \mathrm{U}}{\partial \mathrm{r}} = -30\mathrm{r}^2$
For circular motion of the particle, $\mathrm{F} = \frac{\mathrm{mv}^2}{\mathrm{r}} = 30\mathrm{r}^2$
Substituting $\mathrm{m} = 3\ \mathrm{kg}$ **and** $\mathrm{r} = 10\ \mathrm{m}$,
we get] $\mathrm{v} = 100\ \mathrm{m/s}$
Total energy of the particle in circular motion is
$\mathrm{E} = \frac{1}{2}\mathrm{mv}^2 + \mathrm{U(r)} = \frac{1}{2} \times 3 \times (100)^2 + 10 \times \left(10^3\right) = 2.5 \times 10^4\ \mathrm{J}$
Angular momentum of the particle is $\mathrm{J} = \mathrm{mvr} = 3 \times 100 \times 10 = 3000\ \mathrm{kg\ m^2/s}$

Q.7 Answer:

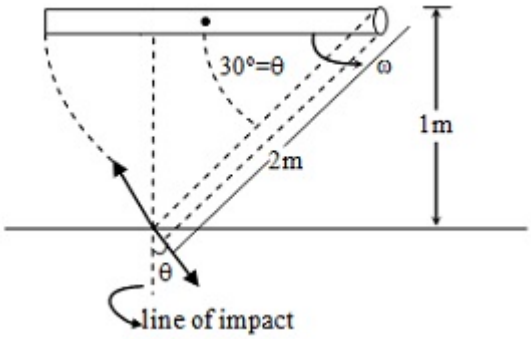
—

Solution:

Q.8 Answer:

—

Solution:



$$mg \times \frac{\ell}{2}\sin 30^\circ = \frac{1}{2} \times \frac{m\ell^2}{3}\omega^2$$
$$\omega = \sqrt{\frac{3g}{2\ell}} = \sqrt{\frac{3g}{4}} = \frac{\sqrt{3g}}{2}$$
$$e = \frac{\omega\ell\cos 30^\circ}{\omega'\ell\cos 30^\circ}$$
$$\omega' = \omega = \frac{\sqrt{3g}}{2}$$

Q.9 Answer:

Force required is $10\ \mathrm{N}$

Solution:

When block is moving with constant velocity, the force F **should be just sufficient to balance limiting frictional force.**
So, $\mathrm{F} = \mu\mathrm{N} = \mu\mathrm{W} = (0.4)(25) = 10\ \mathrm{N}$
Line of action of N **should be shifted a distance** x **from the centre such that algebraic sum of torques about**

@IITJEE_Advanced

G is zero, From figure,

$$-f\frac{H}{2}+F\left(\frac{H}{2}-h\right)+Nx=0$$

$$f=10\ N, H=0.5\ m,\ F=10\ N$$

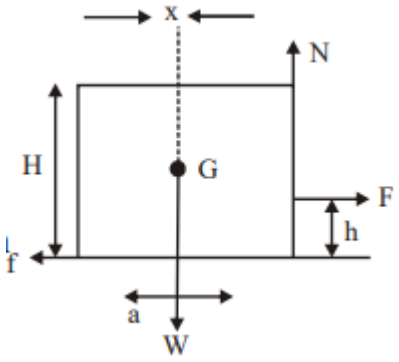
when $h=0.125\ m$ and $N=W=25\ N$

By solving, we get $x=0.05\ m=5\ cm$

For just toppling N should shift to a maximum distance of $x=\frac{a}{2}$ **where a** $0.25\ m$.

So, $-f\frac{H}{2}+F\left(\frac{H}{2}-h\right)+W\left(\frac{a}{2}\right)=0$

By solving, $h=0.31\ m=31\ cm$.



Q.10 Answer:

—

Solution:

Initial angular speed of $0.5\ kg, \omega_0=10 rad/s$.

Due to friction, retarding torque on $0.5\ kg$ **and accelerating torque on** $1.5\ kg$ **till both have common angular speed** ω.

By conservation of angular momentum, $I_1\omega_0=(I_1+I_2)\omega$

$$\therefore \quad \omega=\frac{I_1\omega_0}{I_1+I_2}=\frac{\omega_0}{1+\frac{I_2}{I_1}}=\frac{\omega_0}{1+\frac{1.5}{0.5}}=\frac{\omega_0}{4}=2.5 rad/s$$

Final kinetic energy $K_f=\frac{1}{2}(I_1+I_2)\omega^2$

Loss in K.E. $\Delta K=K_i-K_f=\frac{1}{2}\left[I_1\omega_0^2-(I_1+I_2)\omega^2\right]$

Fraction, $\frac{\Delta K}{K_i}=\frac{I_1\omega_0^2-(I_1+I_2)\omega^2}{I_1\omega_0^2}=1-\frac{I_1+I_2}{I_1}\cdot\frac{\omega^2}{\omega_0^2}$

$$\therefore \quad \frac{\Delta K}{K_i}=1-4\times\frac{2.5^2}{100}=1-\frac{25\times25}{25\times100}=0.75$$

Q.11 Answer:

Kinetic energy of the particle is always conserved

Solution:

$$V=-\frac{1}{r}\Rightarrow U=-\frac{m}{r}\Rightarrow F=-\frac{\partial U}{\partial r}=-\frac{m}{r^2}$$

For (A) : As $\vec{\tau}=\overrightarrow{r}\times\overrightarrow{F}=0$ **: Angular momentum of the particle is always conserved.**

For (B) : Kinetic energy of the particle is not constant due to presence of force (except in circular orbit)

For (C) : The particle's path will depend upon kinetic energy of the particle.

For (D) : As V **depends only on** r **so force is always radial.**

Q.12 Answer:

15π

Solution:

@IITJEE_Advanced

(i) Initial moment of inertia energy of rotation

$\mathrm{I}=$ **moment of inertia of rod** $+$ **moment of inertia of boys**

$=\frac{\mathrm{M}\ell^2}{12}+\Sigma \mathrm{mr}^2=\frac{10\times(2.6)^2}{12}+2\times 25\times(1.3)^2=90.1\ \mathrm{kg\ m}^2$

Initial angular speed $\omega=2\mathrm{n}\pi=2\pi\times\frac{5}{60}=\frac{\pi}{6}\mathrm{rad/s}$

$\therefore$ **Initial angular momentum,** $\mathrm{J}=\mathrm{I}\omega=\frac{270.4}{3}\times\frac{\pi}{6}=15\pi\mathrm{J.s}$

Q.13 Answer:

$15\mathrm{rev/min}$

Solution:

When each boy moves 0.6 **towards the centre, the new distance of each boy from the centre is** $0.7\ \mathrm{m}$,

$\therefore$ **New M.I. of system I'** $=\frac{\mathrm{M}\ell^2}{12}+\Sigma \mathrm{mr}^2=\frac{10\times(2.6)^2}{12}+2\times 25\times(0.7)^2=30.13\ \mathrm{kg\ m}^2$

If ω **' is new angular speed, then according to conservation of angular momentum** $\mathrm{J}=\mathrm{I}\omega=\mathrm{I}'\omega'$

$\therefore\quad \omega'=\frac{\mathrm{I}\omega}{\mathrm{I}'}=\frac{15\pi}{30.13}=\frac{\pi}{2}\mathrm{rad/s};\ 2\pi\mathrm{n}'=\frac{\pi}{2}$

$\mathrm{n}'=\frac{1}{4}\mathrm{rev/sec}=15\mathrm{rev/min}.$

Q.14 Answer:

—

Solution:

(iii) Change in kinetic energy of rotation $=\frac{1}{2}\mathrm{I}'\omega'^2-\frac{1}{2}\mathrm{I}\omega^2=\frac{\mathrm{J}'}{2\mathrm{I}'}-\frac{\mathrm{J}}{2\mathrm{I}}=\frac{(15\pi)^2}{2\times 30.13}-\frac{(15\pi)^2}{2\times 90.1}=24.6\ \mathrm{J}$

Q.15 Answer:

$2\mathrm{R}$

Solution:

Since $u=\sqrt{2\mathrm{gL}}$, **the highest point to which the block shall reach is** $\frac{\mathrm{u}^2}{2\,\mathrm{g}}=\mathrm{L}$ **distance above its initial position.**

Hence at highest point, the thread has rotated by $\frac{\pi}{3}=60^\circ$

and the block is at same horizontal level as centre of cylinder as shown.

$x=\frac{R}{\sin 30^\circ}=2R$

Q.16 Answer:

$\frac{\mathrm{mg}}{2}$

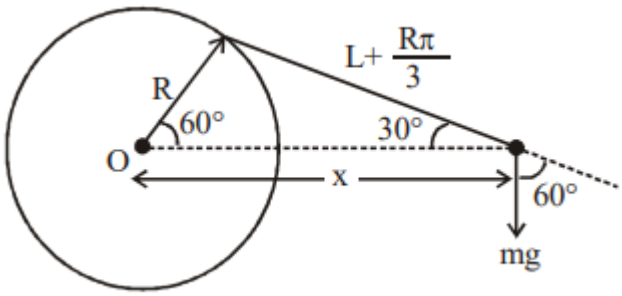
Solution:

$x=\frac{R}{\sin 30^\circ}=2R$

$T=mg\cos 60^\circ=\frac{mg}{2}$

Tangential acceleration, $\mathrm{a_t}=\mathrm{g}\sin 60^\circ=\frac{\sqrt{3}\,\mathrm{g}}{2}$

@IITJEE_Advanced

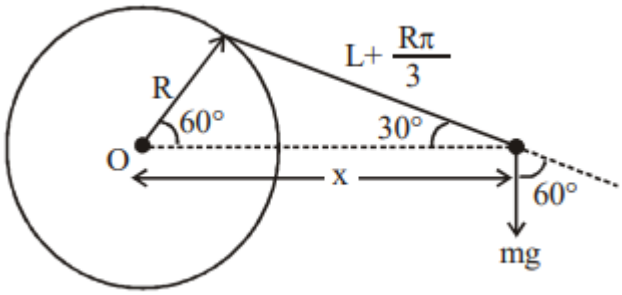


Q.17 Answer:

$\frac{\sqrt{3}}{2}\ g$

Solution:

Tangential acceleration, $a_t = g \sin 60^\circ = \frac{\sqrt{3}\, g}{2}$



Q.18 Answer:

A-(s) B-(s) C-(p) D-(p)

Solution:

$(\mathbf{A}) - \mathbf{s}, (\mathbf{B}) - \mathbf{s}, (\mathbf{C}) - \mathbf{p}, (\mathbf{D}) - \mathbf{p}$

(i) & (ii) $\tau = fR = \frac{mg \sin\theta}{\left(\frac{mR^2}{I}+1\right)}$

If $\frac{mR^2}{I}$ **is minimum t will be maximum.**

$\Rightarrow I$ **is** $mR^2 \Rightarrow \tau$ **is maximum for ring.**

(iii) $v = \sqrt{2\frac{g \sin\theta}{\left(1+\frac{I}{mR^2}\right)}s}$; **v is maximum when** $\frac{I}{mR^2}$ **minimum.** $\Rightarrow$ **v is minimum for solid sphere**

(iv) $t \propto \sqrt{\frac{2\,s}{g \sin\theta}\left(1+\frac{I}{mR^2}\right)}$; **For** t_{min}, $\left(\frac{I}{mr^2}\right)$ **should be min.**

Q.19 Answer:

—

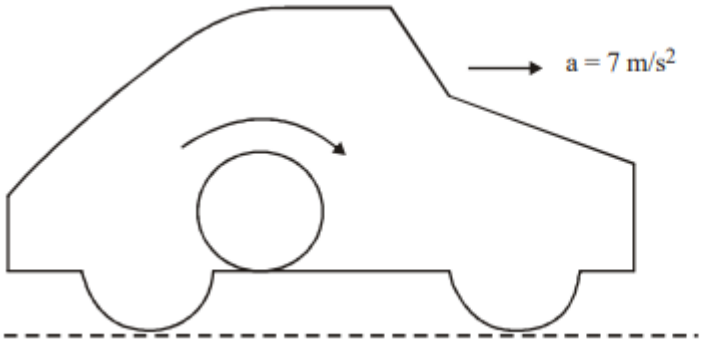
Solution:

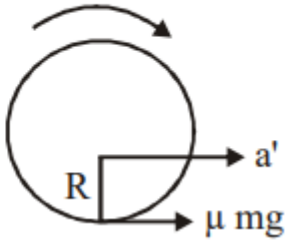
Let a be the acceleration of the car a' be the acceleration of the sphere w.r.t. the man.

Let μ **be the coefficient of static friction between sphere of mass** m **and radius** R, I **be the moment of inertia of the sphere about an axis passing through its centre and** α **be the angular acceleration of the sphere.**

@IITJEE_Advanced

For a man outside, $\mu mg = ma'$ **or** $a' = \mu g$





And $I\alpha = \mu mgR$ **or** $\frac{2}{5}mR^2\alpha = mRa'$

$R\alpha = \frac{5}{2}a' \quad \therefore a = a' + \frac{5}{2}a' = \frac{7}{2}a'$ **or** $a' = \frac{2}{7}a = 2\ ms^{-2}$.

Q.20 Answer:

—

Solution:

The loss of potential energy when the cylinder comes from the top of the track of point $A = mg(H - h) = mg(2.4 - 1.0) = 1.4\ mg$

The gain in its energy $= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$**, where** v **and** ω **are the linear and angular velocities of the cylinder.**

Here, $I = \frac{2}{5}mr^2$ **and** $\omega = v/r$

$\therefore$ **Gain in its K.E.** $= \frac{1}{2}mv^2 + \frac{1}{2} \times \frac{2}{5}mr^2 \times \left(\frac{v}{r}\right)^2$

Now, loss of P.E. $=$ **gain in K.E.**

$1.4\ mg = \frac{7}{10}mv^2; v^2 = \frac{10 \times 1.4 \times 9.8}{7} = 19.6\ m/s$

$v = \sqrt{19.6}\ m/s$

That is the horizontal velocity at point A**.**

The vertical component is zero.

Now time taken to fall a distance $1\ m$ **is** $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 1}{9.8}}$

Distance of position of landing of the ball on ground from B $s = vt = \sqrt{19.6} \times \sqrt{\frac{2 \times 1}{9.8}} = 2\ m$

@IITJEE_Advanced